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Started	Thursday, 29 October 2026, 8:25 PM
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Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It slows down the process
c. It makes it easier to select items based on their relationship to the rest of the items
d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **2**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Searching for data in a list
b. Arranging data in a particular format
c. Deleting data from a list
d. Inserting data into a list

The correct answer is: Arranging data in a particular format

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Insertion Sort
b. Bubble Sort
c. Merge Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Finding a specific element in a list
b. Merging two datasets
c. Removing duplicates from a list
d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **5**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Backtracking
b. Dynamic programming
c. Divide and conquer
d. Greedy method

The correct answer is:
Divide and conquer

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Bubble Sort
b. Merge Sort
c. Quick Sort
d. Heap Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It allows for faster searching
- b. It has no effect on the search operation
- c. It increases the number of comparisons needed
- d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **8**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Searching
- b. Selection
- c. Duplicates
- d. All of the above

The correct answer is: All of the above

Question **9**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is based on the divide-and-conquer approach
- b. It sorts data using a single pass
- c. It is less efficient than bubble sort
- d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **10**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It simplifies the sorting process by using only one pass
b. It avoids the need for recursion
c. It only works on small datasets
d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **11**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Insertion sort
b. Heap sort
c. Quick sort
d. Bubble sort

The correct answer is:
Insertion sort

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Dividing the input into smaller subproblems
c. Comparing each element with the others
d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **13**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Decreases the efficiency of algorithms
- c. Makes data harder to manage
- d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting helps in finding duplicates quickly
- b. Sorting is rarely used in programming
- c. Sorting makes it harder to search for items
- d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **15**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Linear Search
- b. Bubble Sort
- c. Quick Sort
- d. Merge Sort

The correct answer is: Merge Sort